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Digital Twin Construction Method for Monitoring Operation Status of Building Machine Jacking Operation

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Abstract: In the construction of super high-rise buildings, building machines (BMs) are increasingly replacing traditional climbing frames. Building machine jacking operation (BMJO) is a high-difficulty and high-risk stage in the construction of the top mold system. To guarantee the operational safety of the BMJO and to enhance its intelligent control level, a digital twin (DT)-based monitoring method for the operation status of the BMJO is proposed. Firstly, a DT framework for monitoring the operation status of the BMJO is presented, taking into account the operational characteristics of the BM and the requirements of real-time monitoring. The functions of each part are then elaborated in detail. Secondly, the virtual twin model is created using Blender's geometric node group function; artificial neural network technology is used to enable online prediction of the structural performance of the BMJO and a motion model is established to realize a real-time state mapping of the BMJO. Finally, taking a BM project as an example, the DT system is established in conjunction with the project to verify the feasibility of the DT framework for monitoring the state of the BMJO. It is proved that the prediction results have high accuracy and fast analysis speed, thus providing a new way of thinking for monitoring and controlling the safe operation of the BMJO.

Keywords: building machine jacking operation; digital twin; real-time prediction; condition monitoring; Blender

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1. Introduction

1.1. Research Background

High-rise and super high-rise buildings are the trends of urban building development in China. The molding equipment is the key to the construction of super high-rise buildings and structures. Among the various types of molding equipment, the overall steel platform (jacking molding system) and hydraulic climbing mold technology are the most widely employed [1–3]. In the construction of buildings with a height of more than 300 m, there are problems such as large lateral sloshing amplitude, large lateral load, and high safety risk of high-altitude operation. To overcome the difficulties of high-altitude operation, the fourth-generation intelligent integrated platform "air building machine" independently developed by China Construction Third Engineering Bureau Group Co., Ltd. realizes the integration technology of formwork and machinery, which greatly improves work efficiency and construction safety [4–7]. In the process of using the whole construction platform, the jacking operation of the BM represents the pinnacle of construction industrialization, digitalization, and intelligence. Due to the considerable weight of the construction platform and its own mass, the uneven distribution of the load, the variable construction environment, and other factors, it is more challenging to adjust the level of the steel platform during construction and jacking. This places high demands on the attitude control of the jacking system [8]. As the construction height continues to rise, any minor safety hazards may lead to the destabilization and overturning of the entire

steel platform, thus triggering immeasurable serious losses. In contemporary engineering practice, DT technology is becoming increasingly important. This technology uses highly realistic dynamic digital models to simulate and verify the state and behavior of physical entities, aiming to reflect reality with virtual models and control reality through virtual means [9,10]. In this context, the physical entity, virtual space modeling, and virtual-reality fusion interaction mechanism of knowledge modeling methods facilitate the resolution of issues pertaining to the precise control and early safety warning of large-scale equipment operations [11]. In view of the complexity and potential risks in the lifting operation of the BM, it is particularly critical to investigate the DT model of the safety status of BMJO.

1.2. Literature Review

The following studies provide an important theoretical basis for the digital simulation and safety monitoring of the construction process of the BM. Li et al. [12] described the rapid development of aluminum formwork and hydraulic climbing mold construction technology in high-rise buildings and studied the construction and application of the core cylinder under the aluminum formwork + hydraulic climbing mold system, the safety of the frame, and the treatment of special parts. Gong et al. developed new technologies, new devices, and new platforms for man–machine-environment integrated monitoring for key risk sources such as personnel safety management, digital monitoring and control of facilities and equipment, safety status monitoring of integral climbing mold, safety status monitoring of vertical transportation equipment, and construction safety status monitoring and early warning of integral steel platform mold construction site, and a construction safety digital monitoring technology system has been established [13,14]. Through theoretical analysis and finite element calculation, the load transfer system connecting the tower crane and its key large bearing structure has been optimized and improved to reduce the weight and steel consumption of the construction integrated formwork equipment [15]. Zhang et al. [16] emphasized that in the current stage of development, people need to maintain remote monitoring and on-site regulation of the operating status of intelligent construction platforms at all times, and when the platform has a high degree of automation, it only needs to respond to abnormal feedback of the machine and provide support at the necessary moment. Huang et al. [17] proposed an intelligent monitoring system based on PLC + configuration software (The software version number is Siemens WinCC V7.x.)+ sensor integration for real-time monitoring of the integral steel platform, which can provide a reference for improving the safety, monitoring real-time, and automation control of the integral steel platform mold climbing. Zuo et al. [18] statistically and analytically analyzed the key monitoring items and indicators of ICCP (Integral Climbing Construction Platform) during the construction and operation periods and ranked the degree of importance of the monitoring items. Pan et al. have conducted a lot of research on the working state of high-rise BMs. They studied the critical state of BMs in different structural layers and proposed a method based on vibration signal analysis (a three-axis acceleration sensor collects vibration signals) to determine the working state of BM [19]. In the study of dynamic monitoring and early warning research of the BMJO, a series of fiber optic grating horizontal sensors are used to monitor the three-dimensional posture of the BM in real time, and a three-level early warning index and control method is proposed to reflect the safety state during the dynamic lifting process [8]. They employed three multivariate time series neural network models (long short-term memory network, gated recurrent unit, and temporal convolutional network) to predict the operating posture of BM [20]. To address the complex issue of the "human–machine-environment" coordination in the BMJO, Wang et al. [21] developed an intelligent construction application framework for aerial BMJO.

DT technology is widely applied in architectural design, construction, and operations and maintenance. As a key technology for achieving intelligent manufacturing and industrial digital transformation, it plays a significant role in design optimization, construction management, facility maintenance, and emergency response. By leveraging technologies such as the Internet of Things (IoT), artificial intelligence (AI), and cloud computing, DTs en-

able real-time data collection, analysis, and visualization, significantly improving building management efficiency and safety [22]. Additionally, DTs utilize advanced data analytics, such as AI, to process vast amounts of data, enabling status monitoring, prediction, diagnosis, prognosis, and system optimization, thereby enhancing information management and decision-making efficiency in building and asset management. Although currently in its early stages, DT technology has been applied in several large-scale projects, showing great potential. Notably, researchers such as Luo [23], Salem [24], and Jiang [25] have conducted in-depth studies and explorations on the application of DT technology in construction management. For instance, Chacón [26] visualized information from all stakeholders in the construction process as different information channels within their proposed DT dashboard framework, facilitating real-time information sharing and analysis during the construction process. Dai et al. [27] established an IFC semantic model of the construction process, integrating model data and site data into a database to achieve dynamic interaction of twin data, and developed a DT platform for site visualization and control. Liu et al. [28] proposed a Digital Twin-Support Vector Machine (DT-SVM) algorithm to process data stored in virtual models and data collected on-site, enabling intelligent prediction of safety risks in the lifting operations of prefabricated buildings. Szpytko et al. [29] proposed an integrated DT model concept for maintenance decision-making, applying it to cranes in container terminals. Su et al. [30] facilitated the trading of waste materials generated from building demolitions through DT technology. Tao et al. [9,31] proposed a new digital twin-driven PHM (Prognostics and Health Management) method, which can utilize various types of sensors and data processing methods for comprehensive consideration and integration of equipment condition monitoring, fault prediction, maintenance decision-making, etc. Boje et al. [32] indicated that DTs provide a means for synchronizing physical activities with the virtual world. By combining simulation and prediction methods, DTs can offer technical support for dynamic monitoring to ensure safety during the construction process. Zhidchenko et al. [33] constructed an interconnected system methodology and reference framework capable of running a physically based DT for heavy equipment. Zhou et al. [34] proposed a five-dimensional DT model for construction process safety, which can achieve digital representation, parametric monitoring, and automated control of the tower crane jacking construction process. Regarding the establishment of DT models, Liu et al. [35] studied the method for establishing DT models in the construction process of steel structures, including three stages: acquisition and transmission of physical space data, construction of DT virtual models, and model importation. Feng et al. [36] investigated the DT modeling methods for intelligent pump stations, with the model encompassing aspects such as pump station modeling, modeling of the pump station's operational process, and operational analysis modeling. In terms of fault prediction, Dong et al. [37] proposed a method for state assessment and fault prediction of intelligent substation protection systems based on DT technology. This involved the construction of a state evaluation model based on matter-element extension and a fault prediction model based on clustering algorithms, ensuring the safe and stable operation of substation protection system equipment. Talmaki et al. [38] proposed a sensor-based, real-time 3D visualization and geometric proximity monitoring method for articulated construction equipment. This method maps sensor data from the field to the corresponding equipment components in the 3D digital twin, providing real-time sensor updates. DT is a methodology for integrating physical and virtual spaces. Information-physical fusion represents a common challenge in advanced manufacturing strategies such as Industry 4.0, US Industrial Internet, Made in China 2025, and "Internet + Manufacturing". It is a critical scientific issue that must be addressed for the practical application of intelligent manufacturing on the ground [39]. Han [40], combining DT and reverse engineering theories and technologies, proposed an extension of the existing BIM (Building Information Modeling) system to construct an "information-physical" interaction strategy oriented toward DT buildings.

1.3. Research Gaps

The BMJO is the most critical and dangerous phase in the construction process. However, existing research primarily focuses on the monitoring, early warning, and synchronous control of the overall steel platform formwork system [41–49]. While there is extensive research on the concepts, technologies, and application areas of DTs, there is a lack of emphasis on the individual processing methods for physical entities, virtual models, and twin data [50–52]. Additionally, the application of DT technology in BMs remains unexplored. There is significant potential for the development and research of DT technology in the BMJO. DT technology can conduct in-depth analysis in monitoring the BMJOs, enabling functions such as structural analysis, fatigue prediction, and motion prediction of the BM. This technology facilitates predictive control of the equipment and optimization of operating parameters. The DT system for monitoring the operational status of BMJOs integrates DT technology, intelligent algorithms, and the IoT. It features multi-dimensionality, intelligence, high integration, real-time processing, and interactivity. The BMJO involves complex mechanical operations and high structural stresses, where any operational error or equipment failure could lead to severe safety incidents. Therefore, ensuring the safety and stability of the BMJO is paramount. Current research primarily focuses on monitoring and early warning systems for the overall steel platform formwork system and synchronous control during jacking. However, these measures still do not fully encompass all the risks associated with the BMJO. Currently, there has been extensive research on the concepts, technologies, and application areas of DTs, but there is relatively little research on the individual handling methods of physical entities, virtual models, and twin data. Although some researchers have explored the application of DT technology in various functions, its specific application in the BMJO has yet to be fully developed and studied. The application of DT technology in the BMJO has immense potential for development, enabling in-depth analysis to perform structural analysis, fatigue prediction, and motion prediction of the BM, thereby achieving predictive control and optimization of operational parameters.

The reason for adopting DT construction methods is that it not only enables real-time monitoring and control of the BMJO but also enhances the overall safety, efficiency, and intelligence of the construction process through advanced data analysis techniques and predictive models. The multidimensional data and real-time feedback provided by DT technology allow construction managers to more accurately grasp the construction dynamics, promptly adjust and optimize construction plans, and reduce potential risks and resource waste. The DT system for monitoring the operational status of the BMJO integrates DT technology, intelligent algorithms, and the IoT, and it is characterized by its multidimensional, intelligent, highly integrated, real-time, and interactive nature. Through real-time data collection and analysis, it offers more comprehensive safety monitoring and early warning, improving safety and efficiency during the construction process and ensuring the smooth execution of the jacking operation. As construction projects become increasingly complex and involve more high-rise buildings, the application of DT technology will become a crucial means to ensure construction safety and enhance management efficiency.

1.4. Framework Structure

In light of the aforementioned considerations, this paper presents a DT framework for monitoring the operational status of BMJOs in Section 2. The specific construction methods are elucidated in Sections 3–5. Section 3 extracts the semantic information of the physical entities involved in the safe operation of BM jacking, forming a knowledge space representing the safety status of the BM after several mappings and iterative optimization. Section 4 utilizes the Blender platform to create the virtual twin model and constructs a digital asset library of parametric components for the BM. Section 5 focuses on DT interaction. Using the Unity3D platform (The software version number is Unity 2022.x.), it realizes the visualization of the real-time status and predicted posture of the BM during jacking operations. By transmitting the geometric feature information of the physical entities and sensor perception data, real-time status mapping of the BM is achieved. An

artificial neural network is employed to compute the jacking posture of the BM under different working conditions, enabling real-time prediction of the structural performance and motion model of the operational status. Combining the prediction data with control algorithms allows for real-time adjustment of the BM's jacking posture. Section 6 takes an engineering project in Wuhan as an example to demonstrate the functionality of the DT platform and to verify the feasibility of applying DT technology in the BMJOs. This research not only facilitates a more profound comprehension of the operational status of BMJOs but also effectively enhances safety management, thereby ensuring the smooth progress of the construction process and the safety of personnel. The research approach and methodology of this study are illustrated in Figure 1.

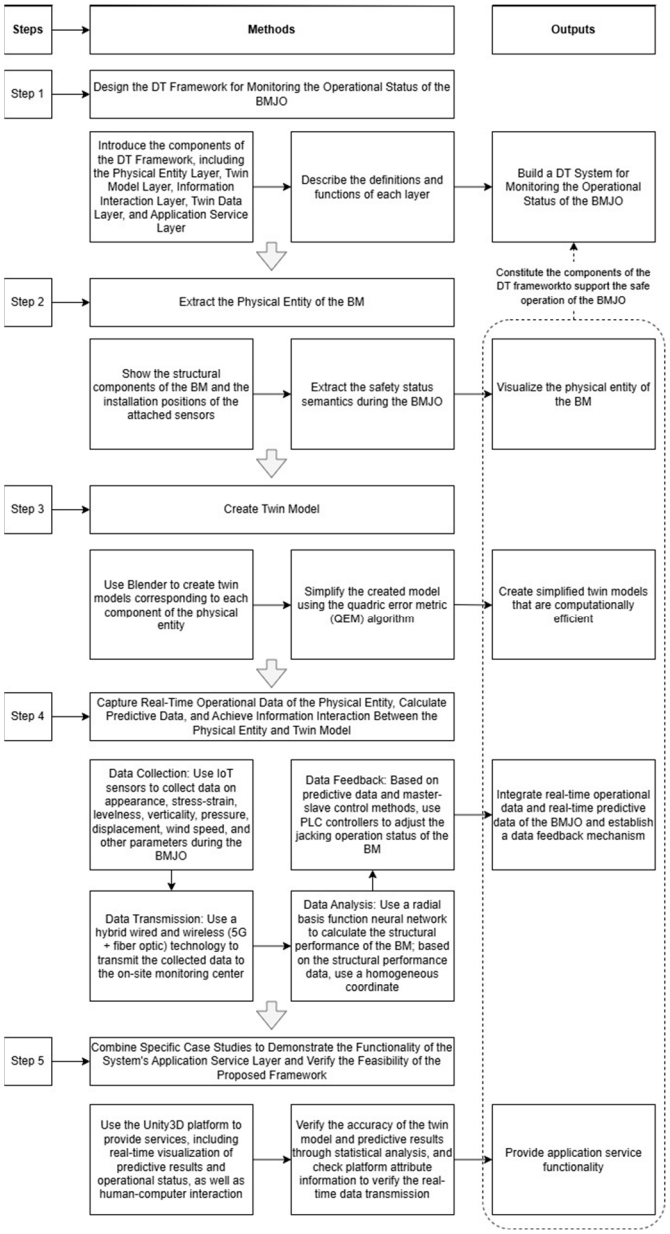


Figure 1. Research approach and methodology.

2. Constructing the DT Framework for Monitoring the Operation Status of BMJO

Based on the five-dimensional DT model proposed by Tao et al. [9,50] and integrating the operational modes and construction statuses of BMs, we construct a DT framework for monitoring the operational status of BMJO. The model structure is illustrated in Figure 2.

This framework establishes a virtual interactive mapping mechanism consisting of the physical entity layer, twin model layer, information transmission layer, twin data layer, and application service layer. This mechanism serves as the foundation for data integration and intelligent monitoring during the construction process, providing technical support for the operational status of BMJO.

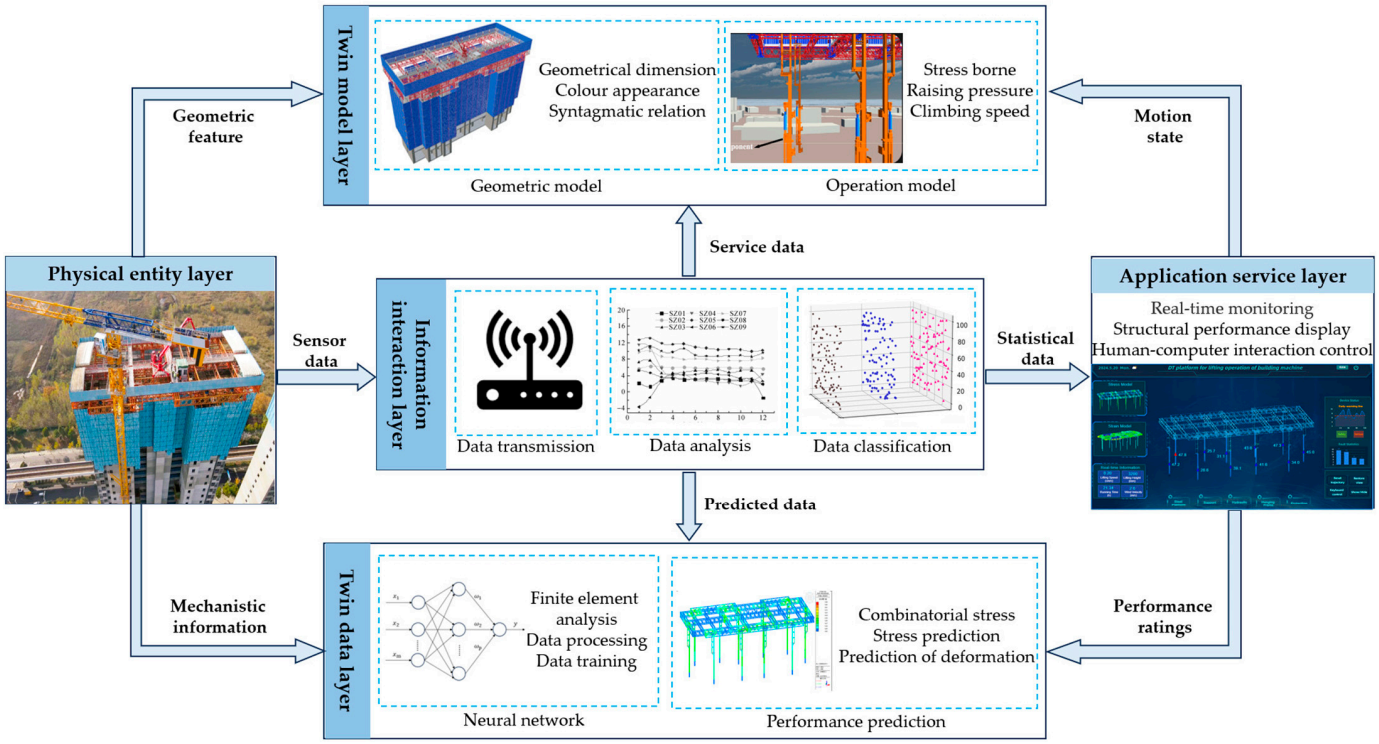


Figure 2. DT system architecture for monitoring the operational status of BMJOs.

2.1. Physical Entity Layer

The physical entity is the foundation of the DT framework and the object of DT model services. It includes the elements related to the jacking operation of the BM, such as personnel, machinery, components, environment, etc., including the BM and sensors placed at various critical locations, which can provide real-time operation data for the entire system.

2.2. Twin Model Layer

The twin model layer is crucial for performance prediction. This part establishes the structural performance prediction model of the BMJO by training the finite element analysis data of the BMJO under different working conditions and establishing the neural network model. Subsequently, the real-time data from the BM sensors are integrated into the prediction model, enabling real-time prediction and attitude control of the operating state of the BMJO.

2.3. Information Interaction Layer

Information interaction is the bridge connecting the real and virtual worlds and is crucial for ensuring the stable operation of the twin system. This layer processes and analyzes the geometric feature information and sensor perception data from the physical entity, storing them in a database. Subsequently, the node data and performance prediction data from the twin model are used as driving data to achieve structural performance prediction and real-time state mapping of the BM.

2.4. Twin Data Layer

The twin data layer is crucial for performance prediction. This part establishes the structural performance prediction model of the BMJO by training the finite element analysis data of the BMJO under different working conditions and establishing the neural network model. Subsequently, the real-time data from the BM sensors are integrated into the prediction model, enabling real-time prediction and attitude control of the operating state of the BMJO.

2.5. Application Service Layer

The application service layer represents the service component of the twin system. It is responsible for integrating data visualization, equipment state monitoring, structural performance prediction, and other functions related to the operation of BM jacking. This layer comprises four principal modules:

1. Status Monitoring Module: This module imports the geometric model of the BM from Blender into the Unity3D rendering engine, where it is combined with the motion model and real-time data to create a virtual representation of the BM. Concurrently, it performs operations such as material processing, texture mapping, and lighting arrangement on the geometric model, enhancing the realism of the virtual BM.
2. Performance Prediction Module: This module utilizes color interpolation algorithms to display the predicted results of the structural performance of the BMJOs in the form of cloud charts. This visualization helps users intuitively perceive the structural performance changes in the BM.
3. Interactive Control Module: This module enables rotation, zooming, and movement of the BM's viewpoint, allowing users to conveniently observe information from different parts of the machine.
4. Data Display Module: This module integrates and statistically analyzes the multi-source data from the BMJOs. It displays key data, such as predictive information and motion states, on the user interface, helping users comprehensively grasp the information pertaining to the BMJO.

3. Semantic Extraction of Physical Entities for Safe Operation of BMJOs

The BM is primarily composed of five principal systems: the steel platform system (or bailey frame system), the hanging frame system, the support system, the hydraulic system, and the safety protection system, as shown in Figure 3. The figure also shows the design and installation positions of the IoT sensors, which are external ancillary entities of the BM.

In order to realize the digital representation of the BMJO in the virtual world, a systematic analysis of key physical entities is conducted. These include the Bailey frame in the steel platform system, the concrete strength of the attached shear wall in the support system, the attached wall support, the guide columns, the support column, and the climbing frame and oil cylinders in the hydraulic system. Based on the operating parameters and corresponding indices of the physical entities, the safety status of the BM's resting state and jacking operations can be determined. In conjunction with the "Standard for assessment and control of safety risk of construction for super high-rise building" (T/CECS 671-2020) [53] and the "Technical standard for self-climbing integrated scaffolding and formwork equipment with steel platform" (JGJ 459-2019) [54], key safety semantics for the resting state and jacking operations of the top mold system are extracted, as shown in Table 1.

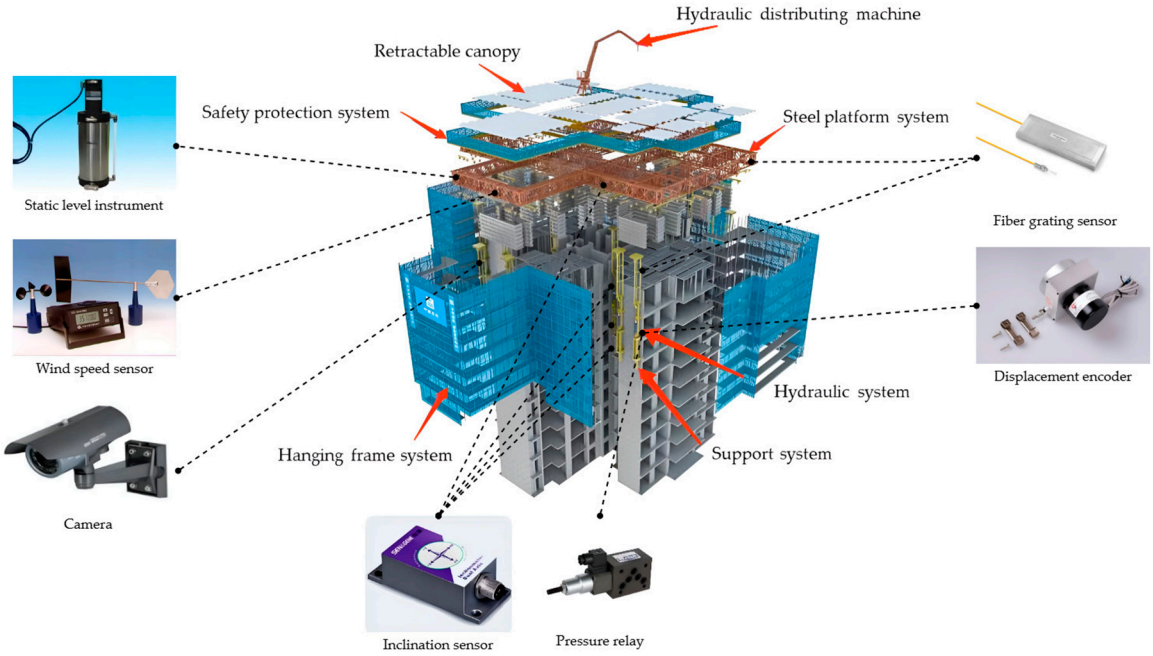


Figure 3. Composition of the BM system and IoT sensor layout.

Table 1. Key semantics of building machine safety state.

Physical Entities	Operation of Physical Projects and Indicators	Importance	Parameter Index	
			Shelving State	Jacking Operation
Steel platform system	Bailey frame level	√√√√√	≤L*/400 or 20 mm	≤L/400 or 20 mm
	Bailey frame acceleration/g	√√√√√	≤±0.0023	≤±0.0023
	Bulk material stacking	√√√√√	Each stacking point according to 500 kg/m^2 control, load limit 16 t	Each stacking point according to 500 kg/m^2 control, load limit 16 t
	Bailey frame stress	√√√	≤ 60% f_s*	≤ 60% f_s
	Bailey frame verticality/‰	√√	1.2	1.2
	Bailey frame vertical deformation	√√	- *	-
	Bailey frame shape attitude	√√	-	-
Support system	Attached shear wall concrete strength/MPa	√√√√√	≥20	≥20
	Wall bearing pressure	√√√√√	≤ 1.2 M_A*/N_S*	-
	Support column displacement/mm	√√√√√	≤5	≤5
	Verticality of guide rail column/‰	√√√√√	1.2	1.2
	Guide rail column stress	√√√	≤ 70% f_r*	≤ 70% f_r
	Column lateral deflection/mm	√√	≤5	≤5
	Wall bearing stress	√√√	≤ 70% f_b*	≤ 70% f_b
Hydraulic system	Cylinder climbing frame verticality/mm	√√√√√	5	5
	Cylinder pressure (synchronous lifting)/MPa	√√√√√	-	≤P_c and pressure difference ≤ 3
	Cylinder displacement (synchronous lifting)/mm	√√√√√	-	Displacement difference ≤ 5
	Verticality of lift cylinder/‰	√√√√√	1.2	1.2
	Climbing speed/(cm·s^{-1})	√√√√√	-	≤0.28
	Lifting displacement	√√	-	0.4 m/time

Table 1. *Cont.*

Physical Entities	Operation of Physical Projects and Indicators	Importance	Parameter Index	
			Shelving State	Jacking Operation
Operating environment	Wind speed/$(m \cdot s^{-1})$	√√√√√	≤32 (8 grade)	≤18 (6 grade)
	Direction	√√√√	-	-
	Illegal operation behavior	√	⊕*	⊕
	Closedness	√	⊕	⊕
	Distance between platform and tower crane	√	⊕	⊕
	Distance between platform and lift	√	⊕	⊕

Note: * indicates the parts in the above table that require additional explanation. L is the maximum width of Bailey frame (mm); f_s is the design value of Bailey frame stress (MPa); f_r is the stress design value (MPa) of the guide rail column; f_b is the stress design value (MPa) of the wall-attached support; M_A is the total load of the platform (t); N_S is the number of fulcrums; P_c is the allowable pressure of the cylinder (MPa); "⊕" is no effective quantitative monitoring method; and "-" are projects that do not involve or do not require quantitative monitoring.

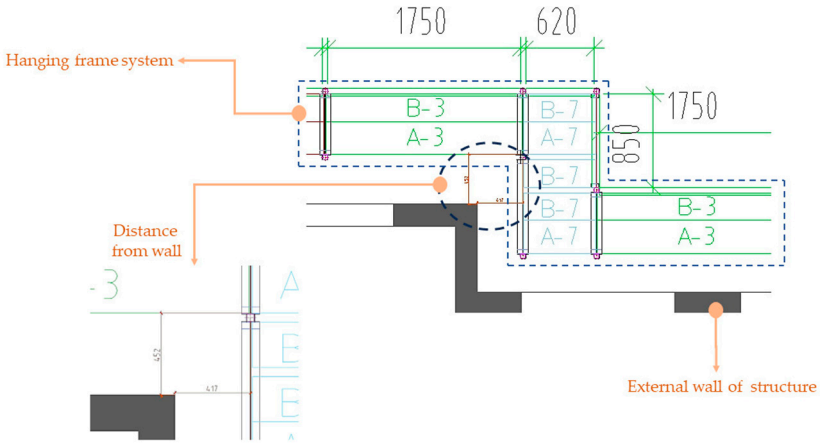
4. Creating Twin Models with Blender

Blender is a free and open-source 3D modeling software (The software version number is Blender 3.4.0.) capable of handling various complex geometric designs and animation requirements. It enables users to create and edit geometry shapes within 3D scenes and to regulate the generation and deformation of these shapes through the use of node diagrams. This nodal approach makes parametric design and geometric processing more intuitive and flexible.

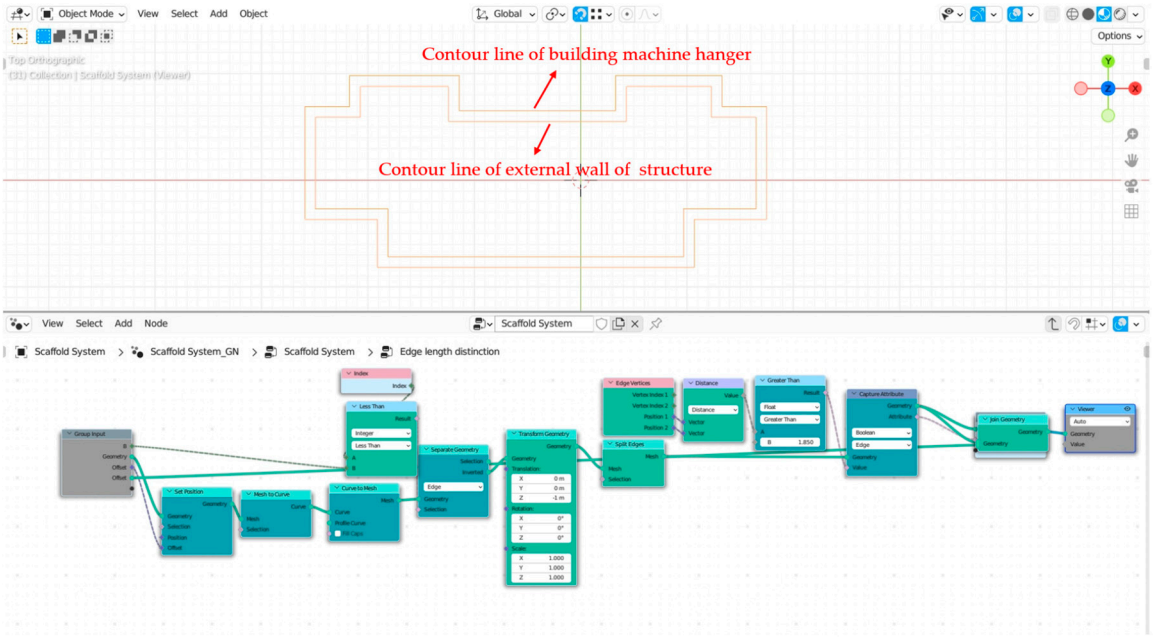
In Blender, the modeling process is carried out by first creating individual components and then combining them into larger assemblies, starting with local parts and progressing to the overall structure. Components with multiple models and sizes are designed in a parametric manner, while other non-parametric components are modeled in a conventional method. The completed models, geometric node groups, materials, and other assets are stored in Blender's digital asset library for easy access and reuse. To create the hanging frame system of the BM, the project's CAD floor plan can be imported into the Blender platform. With the use of visual programming tools and logical rules, a parametric node group can be created that automatically generates the contour line of the hanging frame system based on the building's outline. This process is illustrated in Figure 4 below. Additionally, geometric node groups can be created to automatically generate the components of the hanging frame system. The final result is that by selecting the building's outline, the BM's hanging frame system can be automatically generated. The steel platform system, support system, hydraulic system, and protection system of the BM need to be created manually in conjunction with the structural stress performance and the determined number and position of the pivot points.

For variable components such as protective mesh sheets, walkway plates, connecting beams, boom columns, and other components, parametric modeling is carried out with the geometric node visual programming function in Blender. For variable components such as protective mesh panels, walkway boards, connecting beams, and hanger columns, parametric modeling is performed using Blender's geometric node visual programming functionality. Taking the creation of the hanging frame system's protective mesh panel as an example, as shown in Figure 5, the creation process involves several steps. First, a mesh is added through the mesh node in the mesh primitive, which serves as the mesh surface of the protective mesh sheets. Next, the four attributes of the mesh node are connected to the group input node, allowing for later adjustment of the mesh surface parameters as needed. Finally, instantiate at the point and create a mesh circle centered at the point. Then, extrude the mesh circle along the z-axis to form a mesh cylinder. The mesh surface and mesh cylinders are then combined using a mesh Boolean node to obtain the mesh panel surface. Additionally, the size, material, hole dimensions, and other parameters of the protective mesh panel can be set according to requirements, achieving adaptive modeling of the protective mesh panel. Parametric design of components helps meet the varying

demands of different projects, making the components more flexible and reusable. For components with fixed models, conventional modeling is performed, and these are stored in the digital asset library. Parametric components and non-parametric components are categorized and combined according to the five major systems. Finally, these five systems are assembled together to form a complete visual model of the BM.



(**a**)



(**b**)

Figure 4. Creation of parametric node groups for automatic generation of hanging frame system contour lines: (**a**) Relationship between the external contour of the hanger and the position of the external wall; (**b**) Hanger system contour line generation.

The geometric mesh model of the BM, created using Blender, is composed of triangular faces. Due to the complex structure and large size of the BM, its mesh model contains a large number of triangular faces. The excessive number of triangular faces requires lengthy computation time during subsequent structural performance predictions, which does not meet the real-time requirements of DTs. Therefore, it is necessary to simplify the mesh model to reduce the number of triangular faces without affecting the geometric features of the model. This reduction in the number of triangular faces will decrease the amount of data required for performance prediction and enable real-time structural performance predictions. The Quadric error metric (QEM) edge collapse algorithm is a commonly used

method for mesh model simplification. When performing edge collapses, the algorithm uses the squared distance between the simplified point and the corresponding local region of the original surface as the error metric. It selects the edge with the smallest error for collapse. The basic idea is to repeatedly collapse the edges of the triangular mesh to reduce the number of triangular faces. The folding principle is illustrated in Figure 6.

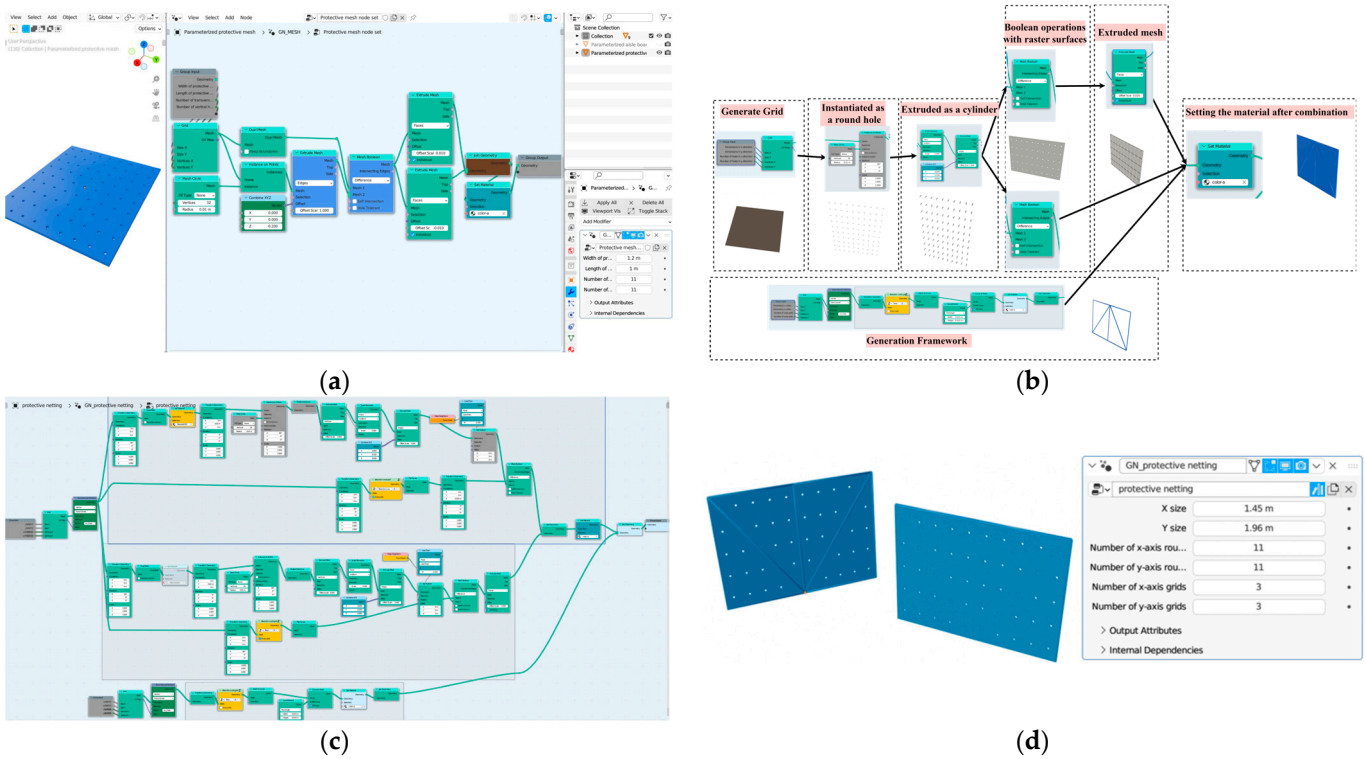


Figure 5. Parametric creation of protective mesh: (**a**) Visual programming process for protective mesh; (**b**) Visual programming details of protective mesh; (**c**) Overview of visual programming for protective mesh; (**d**) Parameterized model and parameterized panel of protective mesh.

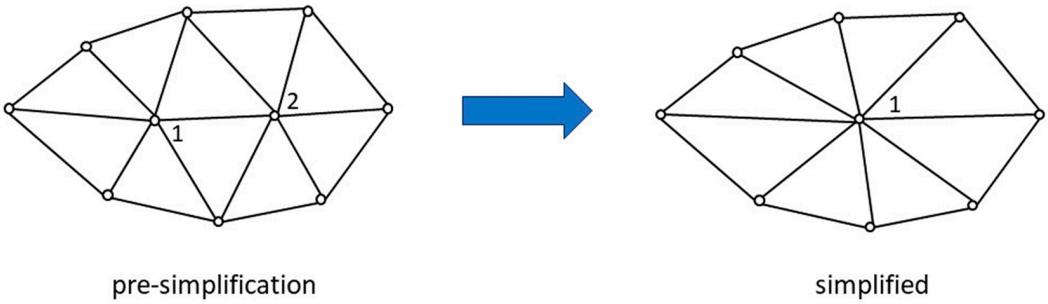


Figure 6. Principle of the QEM edge collapse algorithm.

5. Real-Time Status Mapping and Prediction

5.1. Data Collection and Transmission

The safety of the BMJO is crucial for the secure construction of the top mold system. In conjunction with the key semantics of the safety status of the BM extracted from Table 1, the most crucial monitoring systems during the jacking operation are the steel platform system, the support system, and the hydraulic system. Wind speed is also a significant monitoring factor in the working environment. The monitoring content is composed of the following parameters: apparent monitoring, stress-strain, horizontality, verticality, pressure, displacement, and wind speed. The specific monitoring locations, monitoring roles, names of the instruments used, and images are presented in Table 2 below.

Table 2. Lifting operation data acquisition of BM.

Monitoring Content	Monitoring Site	Monitoring Role	Instruments
Apparent monitoring	Fulcrum guide rail column, wall support	To observe whether the support system is in place, it is required to observe whether the bearing column is correctly placed on the wall support and effectively contacted	Camera
Stress and strain	Fulcrum guide rail column, wall support	Monitor whether the fulcrum position is balanced and deformed	Fiber grating sensor
	Bailey frame	Monitor whether the Bailey frame is balanced and deformed	
Horizontal scallops	Bailey frame	Avoid system offset or instability caused by large elevation difference	Static level instrument
Verticality	Guide rail column	Avoid excessive additional bending moment caused by horizontal displacement, which affects the transmission of vertical load	Inclination sensor
	Bailey frame	Avoid the instability of steel platform caused by horizontal displacement	
	Cylinder climbing frame	Avoid hindering the lifting of the cylinder due to excessive inclination	
	Lifting cylinder	Avoid the influence of excessive inclination on the synchronous lifting of the oil cylinder and the support of the guide rail column	
Pressure	Cylinder upper and lower cavity	Cylinder overpressure, low pressure state	Pressure relay
Displacement	Cylinder plunger	Ensure that each jack-up cylinder rises and falls synchronously	Displacement encoder
Wind speed	Bailey frame	Avoid shaking the entire BM system due to excessive wind pressure	Wind speed sensor

The monitoring scheme for the BMJO employs a local and remote synchronous monitoring strategy, utilizing a distributed monitoring system framework structure, as illustrated in Figure 7. Local monitoring primarily employs fiber optic grating sensors, wireless sensors, and ultrasonic sensors. The sensor monitoring data are transmitted to the data collector equipment via wired and wireless (5G + fiber optic) fusion technology. The collected signals are parsed, formatted, and transmitted to a local industrial computer, with the final data being sent to the on-site monitoring center (Unity3D platform). At the on-site monitoring center, the data are analyzed and processed through an information detection platform and displayed in real-time using the 3D building information model of the BM. This enables staff to quickly view monitoring data from different points, including automated analysis of hydraulic cylinder pressure, wind rose diagrams, structural performance analysis, early warning algorithm analysis, and statistical analysis of values from various measurement points. A fiber optic channel is established between the remote monitoring center and the local monitoring center, enabling the remote transmission, display, monitoring, and operational posture control of the BMJO monitoring data.

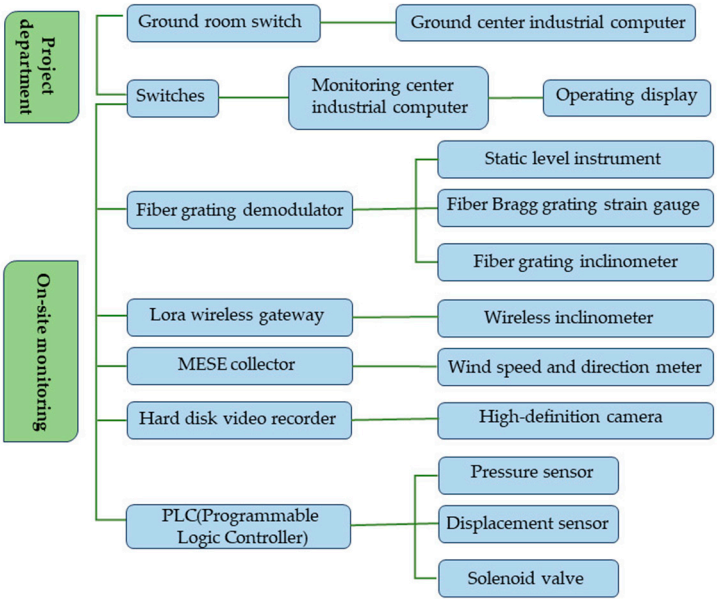


Figure 7. Monitoring System Architecture.

5.2. Real-Time Prediction of BMJOs

In order to establish the dynamic connection between the physical entity and the virtual twin model and realize the real-time prediction of BMJO, the technical route of building a machine jacking operation digital twin (BMJO-DT) system is established, as shown in Figure 8. The construction of the BMJO-DT system is mainly divided into the preprocessing stage and the real-time mapping stage. In the preprocessing stage, the virtual twin model created in Blender is imported into Unity3D using the FBX format. Mesh simplification techniques are employed to reduce the number of meshes, and the simplified mesh model is used as the finite element analysis model. Finite element analysis of the BMJOs is then conducted to obtain the training data required for the agent model. The real-time mapping stage is achieved through the parsing and classification of sensor data on the BM. The virtual twin model, using the motion model, achieves real-time posture mapping of the actual physical entity. At the same time, the predictive model obtained from agent model training is utilized, combined with real-time sensor data, to achieve real-time prediction of the structural performance during the jacking operations.

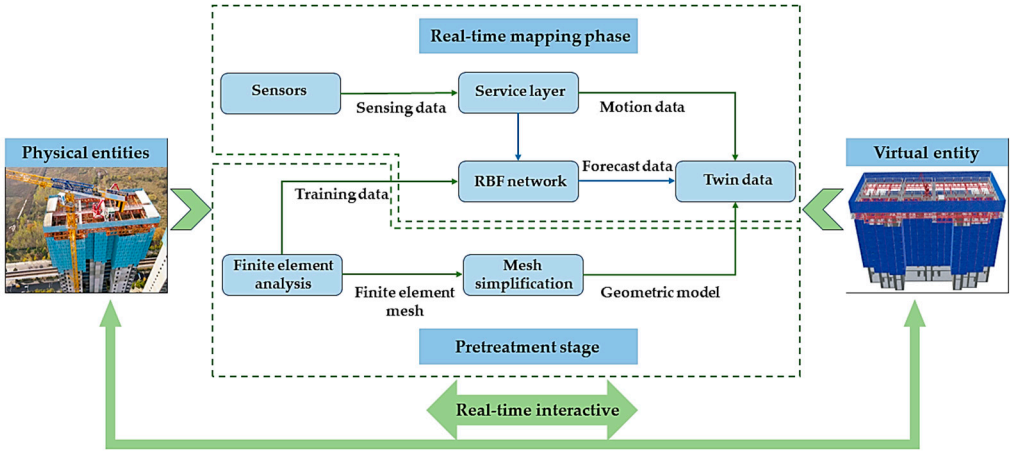


Figure 8. Technical route to a DT system for real-time prediction of BMJOs.

5.2.1. Preprocessing Stage

The BM system is large and structurally complex, making the analysis of its jacking operations time-consuming and costly, which is challenging for real-time computation and online analysis requirements. In order to enhance the efficiency of the solution and reduce the analysis time, an artificial neural network is used to establish a mapping relationship between input state variables and output performance parameters. This approach enables the approximation of the original performance results, thereby facilitating the structural performance prediction of the BMJO under any state. To ensure the accuracy of the prediction results, it is essential to first ensure the accuracy of the finite element analysis. A radial basis function (RBF) neural network is employed to perform finite element analysis of the BMJOs, with the analysis results serving as the training data for the neural network.

The RBF neural network is a type of feedforward neural network that is commonly used to approximate continuous functions. Compared to other types of neural networks, the RBF neural network has the advantages of faster approximation capabilities and better practical usability [55,56]. The RBF neural network consists of three layers: the input layer, the hidden layer, and the output layer. The network structure is illustrated in Figure 9. The first layer is the input layer, which is composed of perception units and is solely used to transmit data without altering the original input data. The second layer is the hidden layer, which employs Gaussian functions as its action functions to perform nonlinear transformations on the input data. The third layer is the output layer, which employs a linear action function, whereby the transformed data of the hidden layer are linearly weighted and outputted, resulting in the prediction result of the neural network.

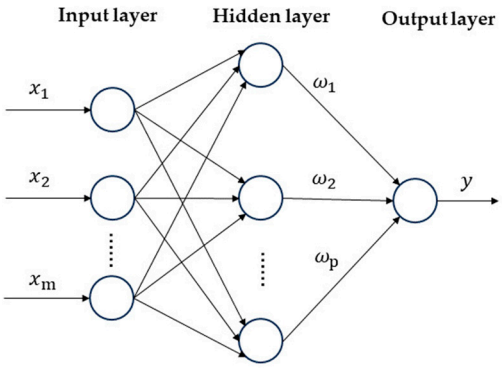


Figure 9. Structure of the RBF neural network.

The activation functions of the nodes in an RBF neural network can take various forms. In this study, the Gaussian function is used as the activation function, which is represented by the following expression:

$$\mathrm{R}(\vec{X}^{p} - \vec{W}^{i}) = \exp(-\frac{1}{2\sigma^2}\|\vec{X}^{p} - \vec{W}^{i}\|^2) \quad (1)$$

where $\vec{X}^{p} = [x_1^p, x_2^p, \cdots, x_k^p]$ represents the p-th input sample and $\vec{W}^{i} = [w_1^i, w_2^i, \cdots, w_k^i]$ is the center vector of the i-th RBF in the hidden layer. Here, $p = 1, 2, \cdots, \mathrm{P}$, where P is the total number of samples.

The output function of the RBF neural network is given by the following:

$$y_j = \sum_{i=1}^{h} w_{ij}^2 \exp(-\frac{1}{2\sigma^2}\|\vec{X}^{p} - \vec{W}^{i}\|^2) \quad (2)$$

where w_{ij}^2 is the weight of the i-th node of the hidden layer to the j-th node of the output layer; $i = 1, 2, \cdots, \mathrm{h}$; y_j is the actual output of the j-th output node of the neural network.

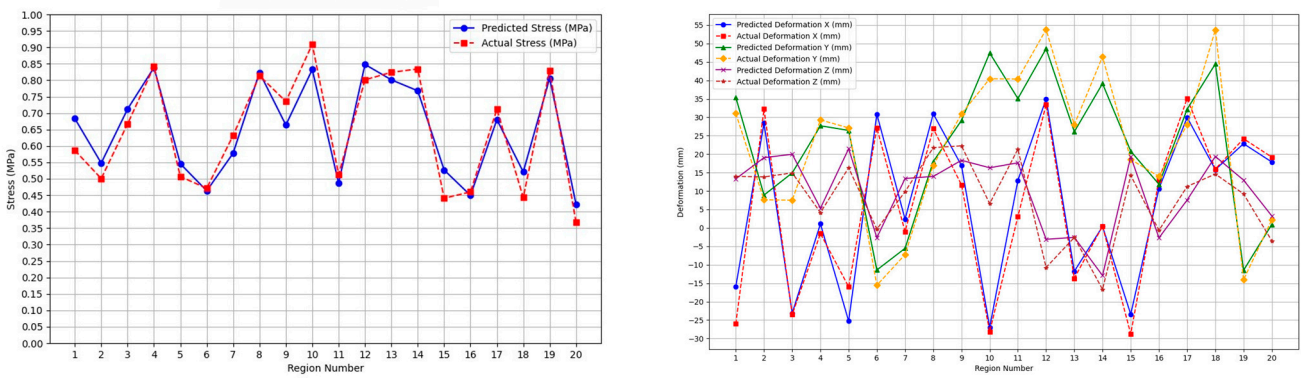
To achieve real-time prediction of the structural mechanical performance of BMJOs, it is necessary to establish a predictive model for the mechanical performance of jacking operations based on the RBF neural network described in Equation (2). The variables under different working conditions are used as input parameters, and the stress and deformation values at each node of the simplified mesh model are used as the output values of the samples. The neural network is then trained to establish a structural performance prediction model for the BMJOs. Firstly, a high-fidelity 3D model of the BM is established in accordance with the physical characteristics of the BM, as described in Section 3. This is followed by the generation of a simplified mesh model through the utilization of mesh simplification technology. Based on the operational characteristics of the BM during jacking operations, appropriate loads are applied (permanent loads: self-weight of the structure, such as steel platform, machined parts, formwork, hanging frame, etc.; live loads: platform stacking loads, hanging frame construction loads, platform construction loads, wind loads, etc.), correct material parameters are selected, and components such as guy wires, tow ropes, steel cables, and bolts are effectively substituted. Finite element analysis is then performed on the BM's jacking operations under different conditions. The main factors affecting the change in the structural mechanical properties of the BM during the jacking operation are material stacking load, wind load, crowd load, jacking height, and cylinder pressure. Therefore, material stacking load, wind load, crowd load, jacking height, and cylinder pressure are used as input variables for the proxy model. Each input variable is confined within the design range of the BM. Since different input variables have different value ranges and quantification standards, significant differences in values may affect the analysis results. Hence, the values of all input variables are normalized, converting dimensional variables into computable scalars, as shown in Table 3. The numerical range indicates the limit range the BM can withstand, the variation range represents the peak activity intervals of each input variable, and the normalized values are calculated based on the variation range intervals.

Table 3. Input variable values and normalization processing.

Input Variable	Value Range	Variation Range	Variation Step Size	Normalized Value
Material Stacking Load (t)	0~16	4~12	0.5	0.25~0.75
Wind Load (KN/m^2)	0~0.20	0.02~0.12	0.02	0.1~0.6
Crowd Load (kg)	0~18750	3750~11250	75	0.2~0.6
Jacking Height (m)	13.6~190.3	13.6~186.3	3.2	0~0.975
Cylinder Pressure (MPa)	0.5~30	0.5	15~25	0.5~0.833

We design multiple sets of different input parameters and use the gradient descent method to train the weight parameters from the hidden layer to the output layer during the prediction process. The centers and widths of the hidden layer nodes are adjusted through iterative learning. We utilize Unity3D to perform simulation modeling of performance indicators for 100 theoretical models generated by randomly combining different levels of the five input variables, resulting in 100 sets of raw data. Using Matlab matrix computation software(The software version number is Matlab R2022b.), a program is written to select 60 sets of data as the training set for multiple iterative calculations. The training of the RBF neural network stops when the training target error reaches 1×10^{-10}, as shown in Figure 10. During the BMJO, the key nodes and their regions of the BM structure are numbered, with a total of 20 key structural performance monitoring areas. These 20 areas will represent the real-time structural performance prediction of the entire BM platform during the jacking operation. We reserve 20 sets of data as a test set for hyperparameter tuning and model selection and the remaining 20 sets of data as a validation set. Validating the model during training helps detect whether the model performs well on the training data but poorly on new data. The calculation method of the RBF neural network breaks through the traditional linear computation model by adopting an integrated synchronous

simulation method. This approach results in shorter calculation steps and more iterations, allowing the simulation result error to be controlled within 10%.



(**a**) (**b**)

Figure 10. Optimized target output prediction curves: (**a**) Predicted structural stress output curve during BMJO; (**b**) Predicted structural deformation output curve during BMJO.

The structural performance prediction of the model on the validation set is evaluated using mean squared error (MSE) and relative error. Table 3 shows the MSE and relative error for the predicted curves of validation set samples for different regional nodes. The MSE is calculated as the square of the difference between the predicted values and the original values. Based on the MSE, the actual error value can be derived, and the ratio of the actual error value to the mean of the original data is taken as the final predicted relative error value. As shown in Table 4, for stress prediction, the relative error is highest in monitoring areas 1 and 4 but remains within 0.1%, with the relative errors in other areas controlled within 0.05%. For deformation prediction, the relative error in the z-axis direction of monitoring area 4 is the highest at 5.56%, remaining below 6%, with the relative errors in other areas controlled within 3%. The prediction accuracy is relatively reliable, allowing for structural performance predictions of various combinations of influencing factors for the BMJO within a reasonable error range.

Table 4. Validation of numerical errors in structural performance prediction.

Monitoring Area	Structural Stress				Structural Deformation				
	Mean Squared Error (MPa)	Actual Error (MPa)	Original Data (MPa)	Relative Error (%)		Mean Squared Error (mm)	Actual Error (mm)	Original Data (mm)	Relative Error (%)
1	0.1849×10^{-6}	0.43×10^{-3}	0.54	0.08	x	0.16	0.4	38	1.05
					y	2.25	1.5	29	5.17
					z	0.04	0.2	−8	2.50
2	0.0441×10^{-6}	0.21×10^{-3}	0.68	0.03	x	0.09	0.3	23	1.30
					y	0.25	0.5	−16	3.13
					z	0.09	0.3	17	1.76
3	0.0324×10^{-6}	0.18×10^{-3}	0.72	0.02	x	0.04	0.2	−15	1.33
					y	0.04	0.2	−11	1.82
					z	0.09	0.3	19	1.58
4	0.0729×10^{-6}	0.27×10^{-3}	0.33	0.08	x	0.81	0.9	24	3.75
					y	0.49	0.7	−25	2.80
					z	1.00	1.0	18	5.56
5	0.1225×10^{-6}	0.35×10^{-3}	0.87	0.04	x	0.00	0.00	13	0.00
					y	0.25	0.5	36	1.39
					z	0.01	0.1	12	0.83
...	...	...	...	...	...	...	...	...	...
20	0.0196×10^{-6}	0.14×10^{-3}	0.49	0.03	x	0.09	0.3	17	1.76
					y	0.36	0.6	23	2.61
					z	0.25	0.5	−14	3.57

Note: Relative errors are taken as absolute values.

5.2.2. Real-Time Mapping Stage

To achieve real-time mapping of the BM's jacking operation posture, a motion model of the BM needs to be established. The initial state of the BM is defined as the vertical direction of its support system being the z-axis, the vertical normal direction of the shear wall attached to the support system being the x-axis, and the direction of the y-axis being derived from the right-handed helix rule. Based on the operational state of the BM during jacking operations, the motion model of the BM is established:

$$P = TP_0 \tag{3}$$

where P denotes the real-time position of each component of the BM; $P_0 = (u_0, v_0, w_0, 1)$ is the homogeneous coordinate vector of the initial state of the BM; T is a homogeneous coordinate transformation matrix, which represents the movement of each component in absolute space, with different components having different transformation matrices. The support system of the BM is fixed on the shear wall by the wall bearing, so the transformation matrix is the unit matrix. The expression of the transformation matrix is the following:

$$T = \begin{bmatrix} R & \overrightarrow{t} \\ \overrightarrow{0}^T & 0 \end{bmatrix} = \begin{bmatrix} cos\,\theta & -sin\,\theta & 0 & u \\ sin\,\theta & cos\,\theta & 0 & v \\ 0 & 0 & 1 & w \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4}$$

The BM is composed of steel components, and therefore, its structure can be considered a three-dimensional elastic body. Under the action of external forces, the elastic body undergoes rotation, positional changes, and shape deformation. In the equation, R is a 3×3 rotation matrix, and $\overrightarrow{t}$ is a 1×3 translation vector. The internal points rotate by an angle θ around the z-axis, and the displacements along the x, y, and z axes are represented by u, v, and w, respectively. These displacements are functions of the coordinates of the points, that is,

$$\begin{cases} u = u(x, y, z) \\ v = v(x, y, z) \\ w = w(x, y, z) \end{cases} \tag{5}$$

According to the theory of elasticity, when considering only small displacements and small deformations and neglecting their second or higher-order terms, the geometric relationship between strain and displacement vectors is as follows:

$$\begin{aligned} [\varepsilon] &= \begin{bmatrix} \varepsilon_x & \varepsilon_y & \varepsilon_z & \gamma_{xy} & \gamma_{yz} & \gamma_{zx} \end{bmatrix}^T \\ &= \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial z} & \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} & \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} & \frac{\partial w}{\partial x} \frac{\partial u}{\partial z} \end{bmatrix}^T \end{aligned} \tag{6}$$

The strain components of a three-dimensional elastic body can be represented by the following matrix:

$$\{\varepsilon\} = \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = \begin{bmatrix} \frac{\partial}{\partial_x} & 0 & 0 \\ 0 & \frac{\partial}{\partial_y} & 0 \\ 0 & 0 & \frac{\partial}{\partial_z} \\ \frac{\partial}{\partial_y} & \frac{\partial}{\partial_x} & 0 \\ 0 & \frac{\partial}{\partial_z} & \frac{\partial}{\partial_y} \\ \frac{\partial}{\partial_z} & 0 & \frac{\partial}{\partial_x} \end{bmatrix} \begin{Bmatrix} u \\ v \\ w \end{Bmatrix} \tag{7}$$

Under the action of external forces, the stress state at any point inside the elastic body is also three-dimensional and can be represented by the column vector:

$$\{\sigma\} = \begin{bmatrix} \sigma_x & \sigma_y & \sigma_z & \tau_{xy} & \tau_{yz} & \tau_{zx} \end{bmatrix}^T \tag{8}$$

Within the elastic range, the physical relationship between stress and strain can be expressed by the matrix equation $\{\sigma\} = [D]\{\sigma\}$, where

$$[D] = \frac{E(1-\mu)}{(1+\mu)(1-2\mu)} \begin{bmatrix} 1 & \frac{\mu}{1-\mu} & \frac{\mu}{1-\mu} & 0 & 0 & 0 \\ \frac{\mu}{1-\mu} & 1 & \frac{\mu}{1-\mu} & 0 & 0 & 0 \\ \frac{\mu}{1-\mu} & \frac{\mu}{1-\mu} & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-2\mu}{2(1-\mu)} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1-2\mu}{2(1-\mu)} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-2\mu}{2(1-\mu)} \end{bmatrix} \quad (9)$$

Here, $[D]$ is the elasticity matrix, which depends entirely on the elastic constants E (elastic modulus) and μ (Poisson's ratio).

Sensors upload the real-time operational data of the BMJOs to the data acquisition equipment. The acquisition equipment parses the received data, categorizes and structures the parsed data, and finally stores the key operational data in the local industrial control computer (ICM).

The DT system transmits a request to the local ICM, which performs a proofreading operation on the request based on the timestamp it carries. Once the proofreading is complete, the ICM returns the real-time data required by the DT system. The DT system predicts the structural performance of the BMJO based on the performance prediction model established in the preprocessing stage and the real-time data. The predicted stress or strain results are displayed in the form of cloud diagrams. Concurrently, the motion state P of the jacking operation is calculated in real time according to Equations (3)–(9), resulting in the motion model of the virtual BMJOs. This enables the virtual–real state mapping of the BMJO.

5.3. Synchronization Control of Jacking Operations

The BMJO is not synchronized due to the uneven load distribution of the frame structure. If the jacking synchronization is poor, it will inevitably lead to the deformation of the jacking frame system, which will bring great risks to the construction process. The synchronization control of the jacking system is crucial for monitoring and controlling BMJO. According to the real-time monitoring data and real-time mapping data of the above-mentioned sensors, the operational posture of the BMJO is controlled.

The "primary–subordinate mode" is determined as the control method for the synchronized jacking of the formwork system. The "primary–subordinate method" is to designate one operation element as the primary element and give the primary element a desired output value, and the other elements will follow the action of the primary element with the actual output of the primary element as their desired output value to achieve the synchronization effect. Specifically, in the jacking operations of the BM, one of the multiple hydraulic cylinders is used as the primary cylinder, while the others are subordinate cylinders. By adjusting the opening gain of the proportional directional valve of the primary cylinder to set the jacking speed, the subordinate cylinders follow the movement with the displacement of the primary cylinder as the reference. It has been demonstrated that the disparity in cylinder synchronization displacement resulting from disparate cylinder loads is the most pronounced [57]. Therefore, the heaviest-loaded cylinder is designated as the primary cylinder, and the other cylinders follow the primary cylinder during jacking. In comparison to the "parallel method," which is characterized by a straightforward control principle, high performance expectations for control components, and a lack of resilience to interference, the "primary–subordinate method" is more adept at enhancing control precision in the context of the complex and dynamic environmental conditions encountered during the jacking of the top mold system.

The controller incorporates a programmable logic controller (PLC), which includes an input/output (I/O) module. The indicators for real-time monitoring, evaluation, and logic

control include cylinder jacking displacement, cylinder pressure, and cylinder displacement difference. The synchronization principle for the BMJO is shown in Figure 11. During the BM jacking process, a jacking command is issued from the control console, causing the hydraulic components, such as the motor and oil pump, to start working and the cylinders to begin moving. The displacement sensor on each subordinate cylinder collects the displacement in real time and inputs it into the PLC controller as feedback, which compares it with the real-time displacement of the primary cylinder to calculate the displacement error. The PLC controller utilizes the displacement error and the fuzzy PID (proportional-integral-derivative) algorithm to compute the controller output [58]. The output control signal is sent to each subordinate cylinder, adjusting the size of the hydraulic proportional directional valve openings to control the flow rate and speed, thereby tuning the jacking speed of the cylinders. This allows the subordinate cylinders to move toward the trend of reducing the displacement error with the active cylinder, thus enabling synchronous control.

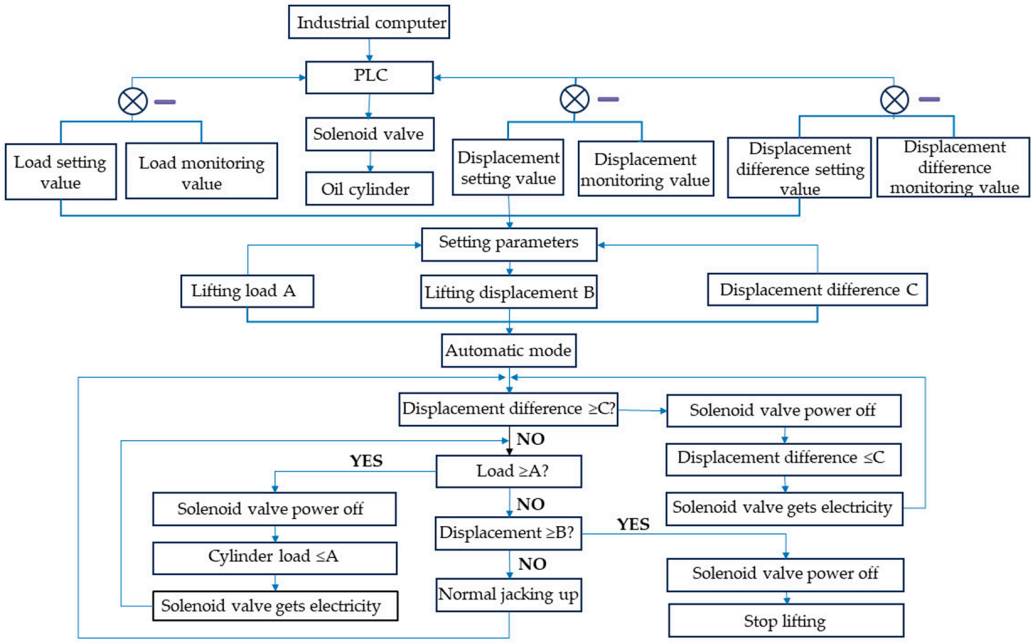


Figure 11. Principle of synchronized control for BM jacking.

6. Platform Functions and Verification Analysis

With the use of the framework proposed in this paper, a DT platform for monitoring the operational status of BMJOs was built with the Unity3D platform for Building 5 of a reconstruction housing project in Wuhan, China. This building has two underground floors and sixty-five above-ground floors, with a total height of 186.3 m. The construction of the core cylinder structure was completed using a fourth-generation BM developed by China Construction Third Bureau Group Co., Ltd. (Headquartered in Wuhan, China). The support and jacking system includes a total of 11 support points, and the jacking operation of the BM is completed by the power pump station, the jacking cylinder group, the synchronous control cabinet, and monitoring equipment. In terms of structure, the height and complexity of Building 5's structure make it suitable for testing and demonstrating the advantages and application effects of DT technology. Technologically, this building uses the fourth-generation BM, equipped with an advanced support and jacking system, providing excellent technical conditions and a data foundation for the construction of the DT platform. Regarding data availability, Building 5 is equipped with comprehensive monitoring devices during construction and is capable of real-time data collection, which ensures reliable data acquisition and real-time monitoring for the DT platform. In terms of project progress and coordination, the construction progress of Building 5 aligns better with the research requirements, and the project team has shown a high level of cooperation

and support for DT technology. These factors ensure the scientific, representative, and operational feasibility of the research, enabling the study to proceed smoothly. The site photos of the construction and completion of the project are shown in Figure 12.



(**a**) (**b**)

Figure 12. Project site photos: (**a**) Building 5 construction photo; (**b**) Building 5 completion photo.

6.1. Introduction to DT Platform Functions

The constructed DT platform, as shown in Figure 13, comprises three principal functional modules: real-time prediction of structural performance and movement status of the BM, real-time display of operation data, and human–computer interaction.

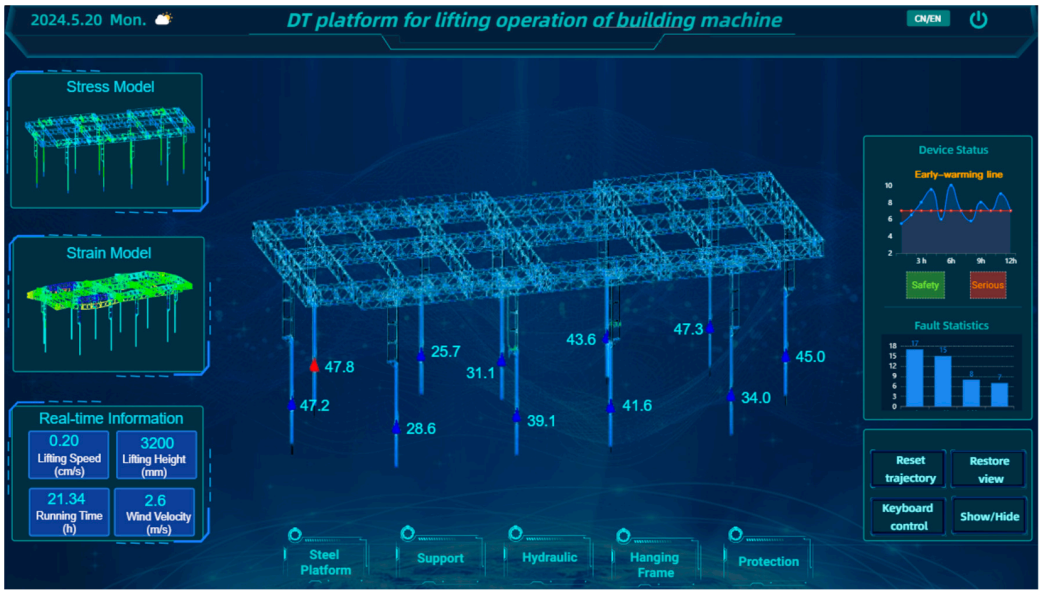
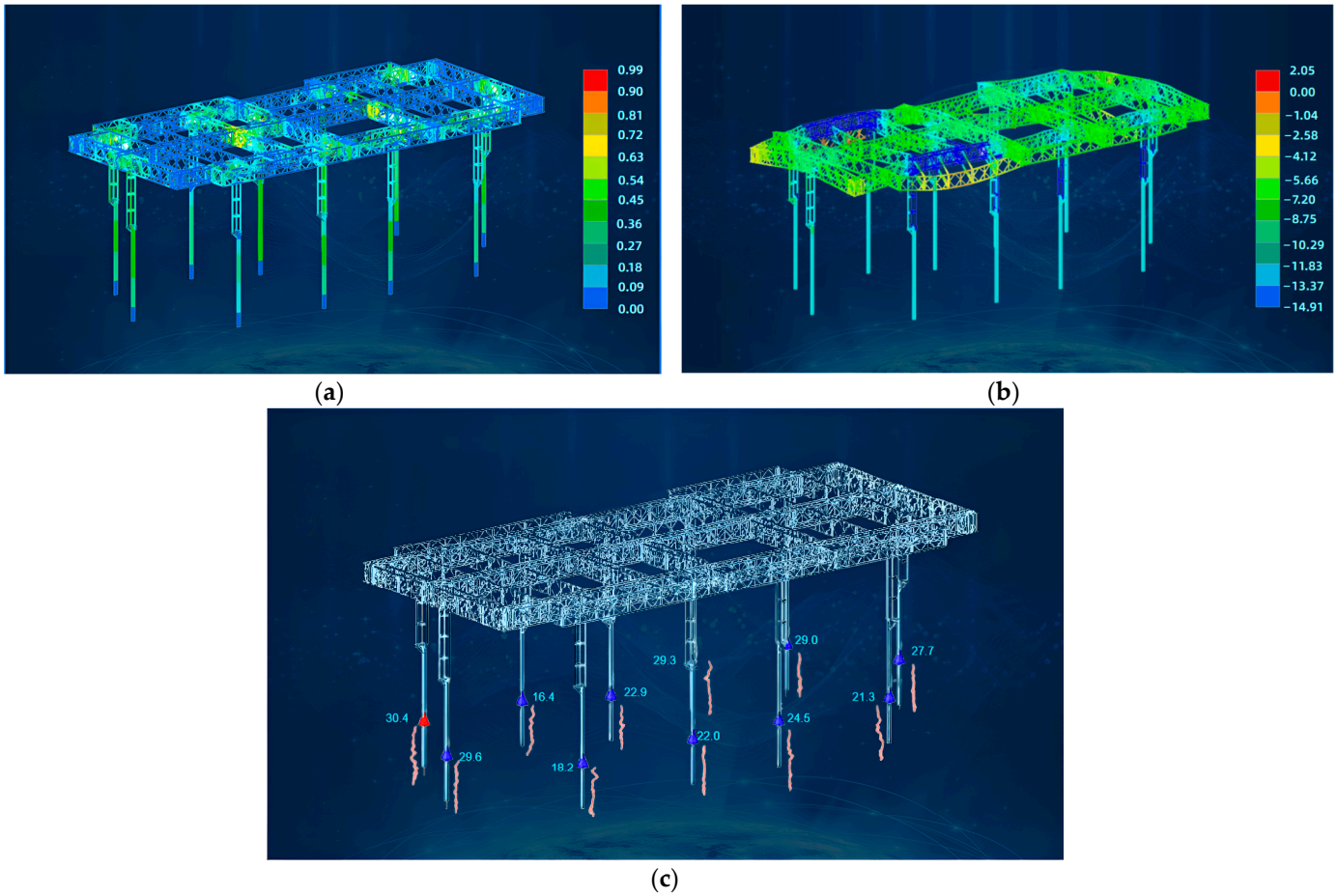


Figure 13. DT platform for the operational status of BMJOs.

6.1.1. Visualization of BM Structural Performance

The visualization of the structural performance of the BM is the core of the DT platform. This module normalizes the real-time predicted performance data and uses a color interpolation algorithm to display the predicted performance values as color changes, allowing users to intuitively perceive the structural performance variations in different parts of the BM. Additionally, the current jacking operation motion state of the BM is displayed in real-time on the monitor using the motion model, and the jacking trajectory of the BM is also shown on the screen using lines, as illustrated in Figure 14.



(**a**) (**b**)

(**c**)

Figure 14. Visualization of structural performance in BMJOs: (**a**) Structural deformation prediction; (**b**) Structural stress prediction; (**c**) Lifting trajectory display.

6.1.2. Real-Time Data Display

The data display module presents real-time information on the BMJOs, including working data and statistical data, on the monitor. This allows users to fully understand the operational posture of the BM during jacking operations and improve jacking efficiency. Specifically, the data display module includes information on the change of movement, such as material stacking load, crowd load, jacking speed, jacking height, cylinder pressure, and so forth. This information is presented in Figure 15. Additionally, the module includes the maximum and minimum performance values of the major structural components during jacking operations, as well as historical data statistics of the mechanical performance of the BM's structural component.

6.1.3. Human–Computer Interaction Control

The human–computer interaction control module represents a further division of the specific functions of the DT system. This module facilitates the delivery of key information to users in a more convenient manner. Specifically, it includes the following functions: the interaction of the BM model, which can rotate and zoom the BM to view the performance of the BM from different perspectives; the on-site jacking status can be transmitted to the platform in real time through the monitoring equipment to realize the real-time mapping between the twin model and the on-site conditions, as shown in Figure 16; and the function of performance selection, which can choose to display the stress or strain data. The system allows the user to select the entire equipment system of the BM, or alternatively, to predict the performance of specific structural components of the BM. It also enables the user to predict the performance of the BM in its entirety. Furthermore, the system allows

the user to modify the scale range of the performance cloud map of the BM structure, with the structural performance cloud map of the BM being readjusted according to the modified range.

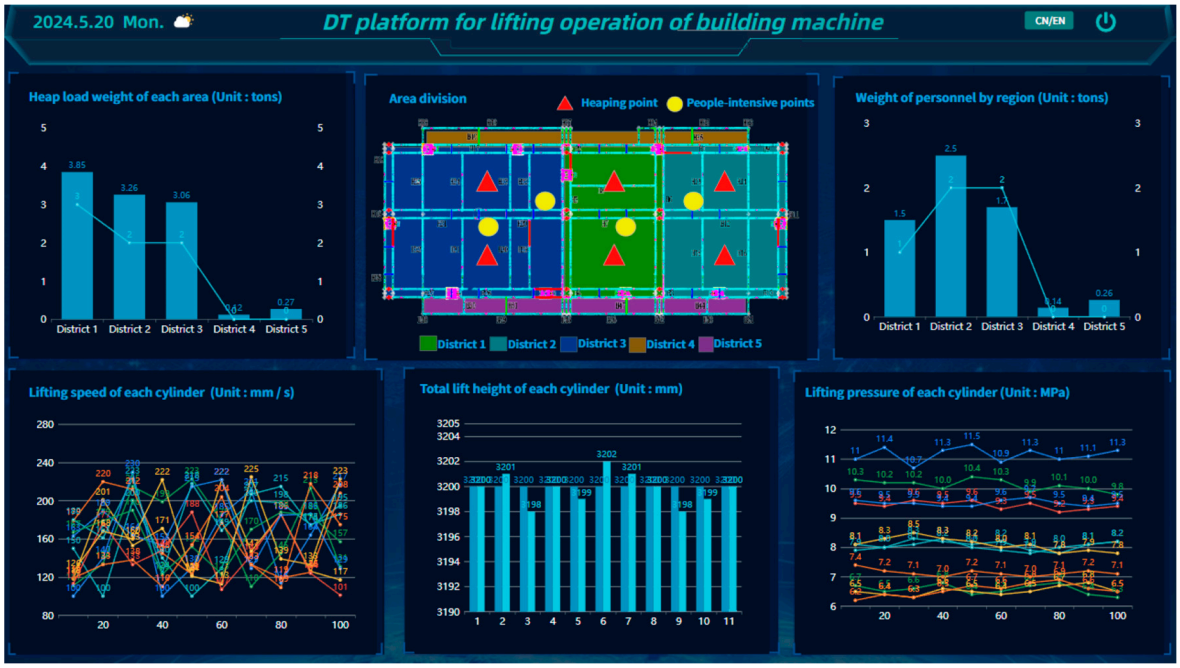


Figure 15. Real-time data display for BMJOs.

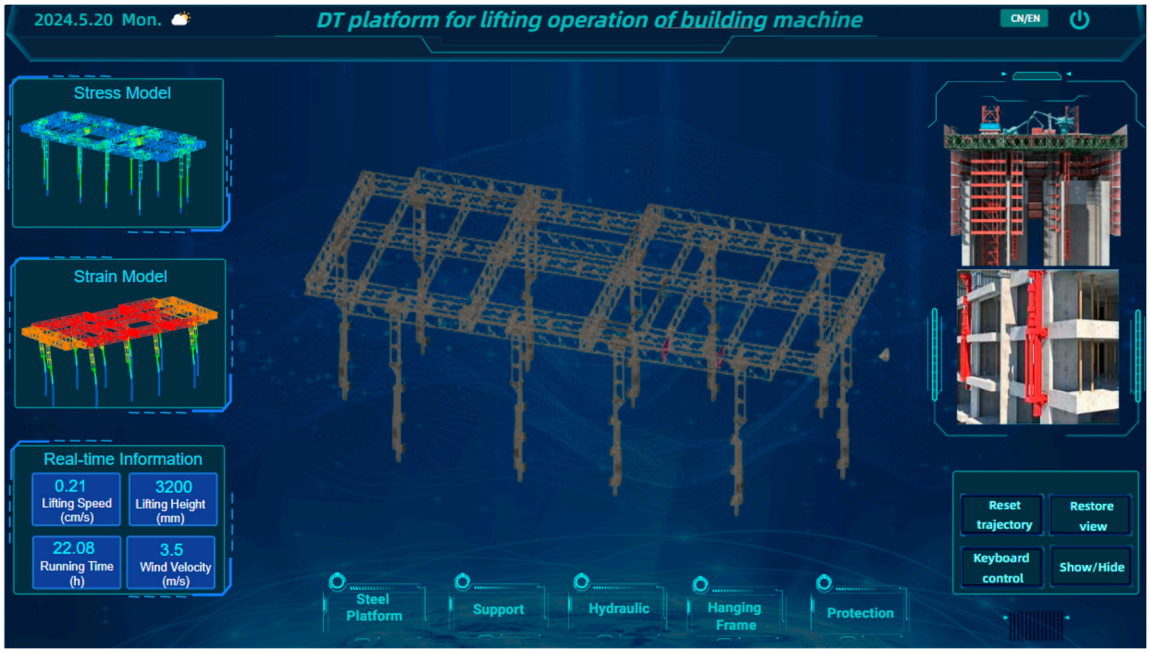


Figure 16. Real-time mapping of BMJOs.

6.2. Verification Analysis

6.2.1. Results of Creating Twin Models with Blender

The BM designed for Building 5 of the project consists of 5 major systems, 3627 components, and 13,000 parameters. The digital asset library of the BM components created in Blender is shown in Figure 17. In Section 5, the generation of the hanging frame system was achieved through the creation of geometric node groups, allowing for one-click generation

of parametric components. It is necessary to verify the accuracy of this auto-generation process and to sample the number of auto-generated components of the hanger system. The resulting statistical results are presented in Table 5 below. The remaining systems (steel platform system, support system, hydraulic system, and protection system) are manually arranged using components from the digital asset library. Errors in these systems are considered human errors and are not statistically analyzed. For the sampled components, the total number of components required was 658, and the number of automatically generated components was 648. The maximum error in automatically generating a single component type was four components, while the minimum error was 0 components. The total accuracy rate was 98.48%, and the error met the accuracy requirements for automatic generation of components. Creating the virtual twin model of the construction machine in Blender and generating parametric construction machine components significantly reduces the time designers need to create models from scratch for each project, enhancing the reusability and flexibility of the models.

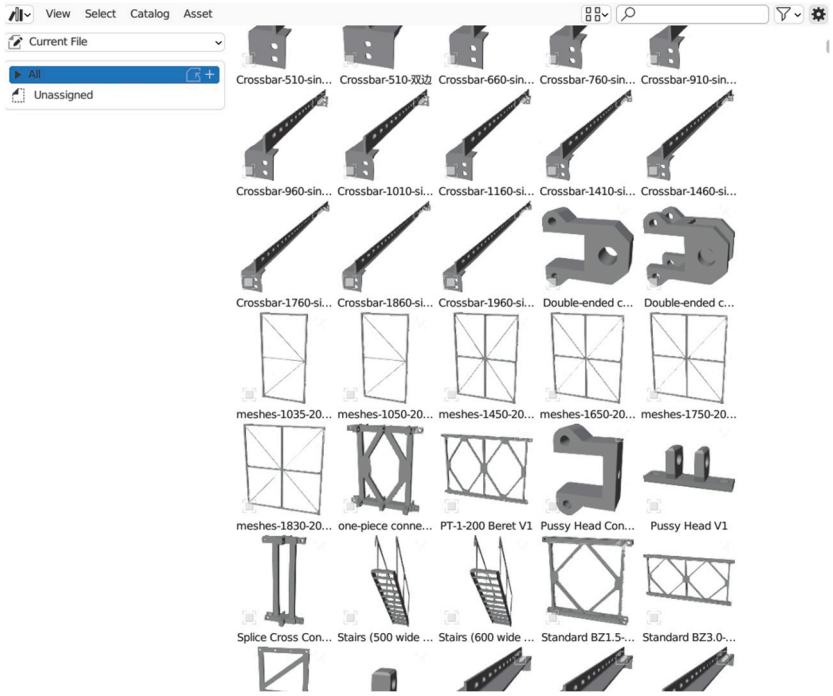


Figure 17. Digital asset library for BM components.

Table 5. Sampling inspection of the number of hanging frame system components.

Component Name	Model	Size	Actual Quantity	Automatic Number Generation
Aisle board *	A-1	1824 × 300	15	12
	A-6	1790 × 300	51	51
	B-2	660 × 250	29	29
	B-3	1300 × 250	53	53
	C-2	1490 × 200	48	48
	C-4	1790 × 200	25	23

Table 5. *Cont.*

Component Name	Model	Size	Actual Quantity	Automatic Number Generation
Protective mesh *	FHW-03	1950 × 1495	160	160
	FHW-06	1730 × 1495	15	14
	FHW-08	1580 × 1495	22	22
	FHW-10	1450 × 1495	190	186
	FHW-13	1260 × 1495	50	50

Note: * indicates the parts in the above table that require additional explanation. There are 22 types of aisle boards, A1~A8; B1~B8; C1~C6. There are 14 types of protective mesh, FHW-01~FHW-14.

6.2.2. Comparison of Performance Prediction Results

To validate the accuracy of the BMJO prediction model, Figure 18 shows the relative error between the finite element analysis results and the prediction model results under 100 different conditions of material stacking load, wind load, crowd load, jacking height, and cylinder pressure. The figure demonstrates a high consistency between the prediction model and the simulation results.

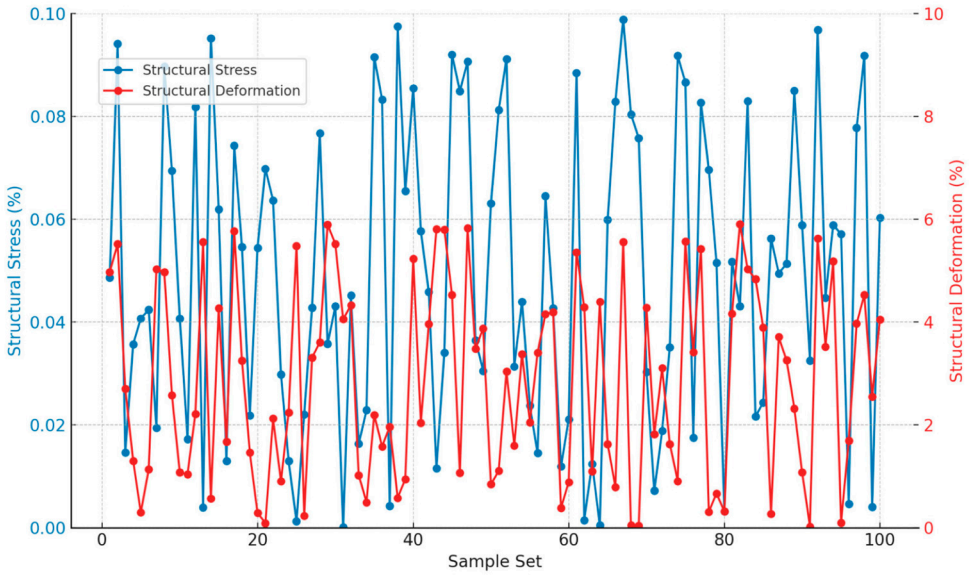


Figure 18. Error between finite element analysis results and prediction results.

6.2.3. Real-Time Verification

Unity3D software (The software version number is Unity 2022.x.) includes a built-in performance analysis tool that allows for monitoring CPU, GPU, and rendering time usage. The test PC for this system is configured with an Intel i7-6700h CPU and a GTX 1060 3 GB GPU. The analysis results from the Unity3D profiler are shown in Figure 19. The maximum time taken for the computer to process one set of data is 151.03 ms, demonstrating that the DT system can perform real-time prediction of the structural performance during the BMJOs, thus meeting the real-time requirements of the DT system.

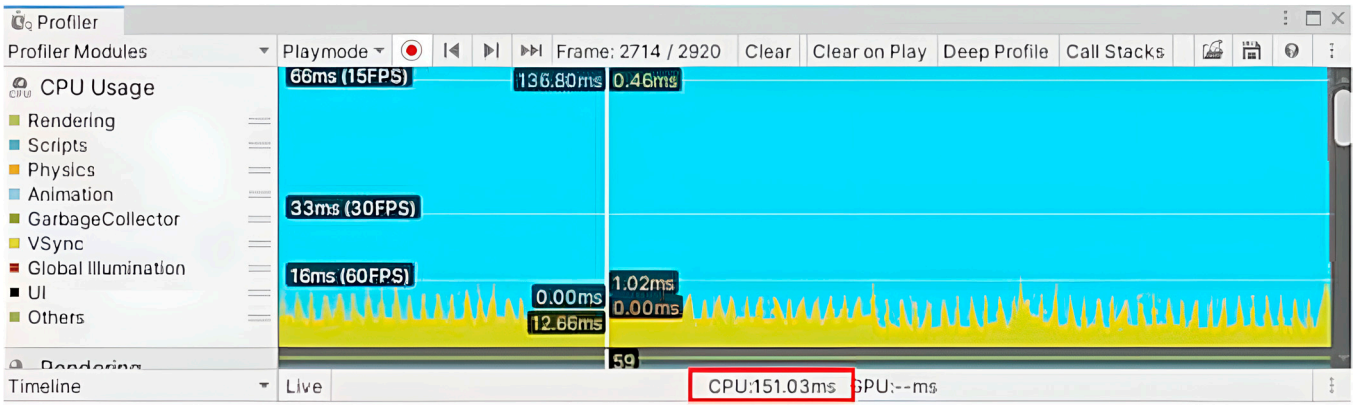


Figure 19. Real-time structural performance analysis in Unity3D.

7. Results

Compared to existing systems that separate monitoring and control, the DT system proposed in this study offers significant advantages in monitoring, evaluation, early warning, and control. By comparing this system with similar research in related fields, our system shows marked improvements in prediction accuracy, computation speed, and intelligence levels. For example, in the study by Liu et al. [28] on safety risk prediction for prefabricated component lifting, traditional monitoring systems can achieve basic status monitoring but have shortcomings in data processing speed and prediction accuracy. In contrast, this study utilizes advanced artificial neural networks and virtual twin technology, overcoming these limitations and significantly enhancing the overall performance and application value of the system.

7.1. Safety and Efficiency

The DT-based monitoring system for the operational status of BMJOs constructed in this study significantly improves the safety and efficiency of the BMJO process in practical applications. The system's real-time monitoring, evaluation, and early warning functions can promptly identify potential hazards and take preventive measures, thereby reducing the likelihood of accidents. Additionally, the online prediction of structural performance and real-time mapping of operational posture make the construction process more transparent and controllable, enhancing the stability and safety of the construction.

During the construction process of the aforementioned project, the DT platform monitored the stress conditions and displacement data at the 11 support points and key node areas of the BM in real time. During the jacking operation, if the system detected abnormal stress at a support point, it would immediately issue a warning signal, prompting the construction personnel to pause operations, inspect, and adjust the equipment to avoid potential safety accidents. Data show that through real-time monitoring and warning functions, the project avoided five major safety accidents during construction, ensuring the safety of construction personnel. Using artificial neural network technology, the system performed online predictions of the structural performance of the BMJO. The system predicted the stress distribution of the cylinder groups during the jacking operation and compared it with actual monitoring data, with the prediction results having an error margin within 0.1%, demonstrating high accuracy. Through online predictions, the system helped the construction team identify and resolve three structural performance anomalies in advance, improving the stability and safety of the construction.

The application of the DT platform reduced the on-site inspection time for engineers before jacking, optimized the equipment adjustment and configuration process, and reduced waiting times and communication costs. Data indicate that the overall efficiency of the jacking operation increased by approximately 19.75%. Specifically, the average time for a single jacking operation decreased from 3 h to 2.4 h. The significant acceleration in construction progress was also due to the system's ability to display operational data

and status in real time, allowing construction personnel to make faster decisions and adjustments, further enhancing construction efficiency.

7.2. Cost–Benefit Analysis

The system brings significant economic benefits to project stakeholders (including construction parties, investors, and regulatory authorities) by reducing accident rates and improving construction efficiency. Although initial development and deployment of the system require investment, the cost savings from reduced accidents and increased efficiency far exceed the initial outlay, demonstrating a strong return on investment.

For the construction party, improved construction efficiency shortened the overall project duration by 10%, equivalent to a reduction of approximately 2.4 months from the original 24-month schedule. This not only lowered on-site management and labor costs but also reduced construction machinery rental expenses. Statistics show that the construction party saved a total of 5.025 million RMB, including 962,700 RMB in on-site management fees, 2.2135 million RMB in labor costs, and 1.8788 million RMB in equipment rental fees. For the investors, the project's early completion allows them to put the property into use sooner and start generating income earlier. With a monthly rental income of 1 million RMB, the 2.4-month advance in project completion results in an additional revenue of 2.4 million RMB. Additionally, the application of the DT platform increases the project's competitiveness and attractiveness in the market, helping to enhance the investor's brand value and market reputation. For regulatory authorities, the system's real-time monitoring and data display functions provide detailed construction data and status information, allowing regulators to understand project progress and safety conditions in real time. This improves regulatory efficiency and accuracy. By reducing the occurrence of safety incidents, the system helps to lower regulatory costs and resource input, enhancing overall regulatory effectiveness.

7.3. Production Planning and Construction Dynamics

Through real-time monitoring and online prediction, the system provides construction managers with detailed operational status and performance data, optimizing construction planning and resource allocation. This data-driven management approach enhances the flexibility and responsiveness of construction organizations, making the construction process more efficient and smooth. Throughout the entire project, a total of 61 jacking operations were performed, with the system detecting and addressing 12 instances of abnormal stress conditions, ensuring the smooth execution of all jacking operations. Additionally, the system's remote control functionality offers new methods for intelligent site management, contributing to refined management and intelligent decision-making on construction sites.

In summary, the DT-based monitoring system for the operational status of BMJOs is significant in enhancing construction safety, efficiency, and intelligence. It provides new ideas and methods for future research and practice in related fields. Analysis of specific application examples demonstrates that the DT-based monitoring system for BMJO has significant practical importance and application value in improving the safety and efficiency of BMJO, bringing substantial economic benefits to project stakeholders, and enabling intelligent planning of production schedules and construction dynamics. This not only provides strong assurance for the successful implementation of this project but also offers valuable experience and reference for the development of similar projects in the future.

8. Conclusions

To achieve digital risk management during BMJOs and to improve the efficiency, smoothness, and safety of these operations, it is necessary to implement the following measures. This paper begins by constructing a DT system for monitoring the operational status of BM jacking. It details the contents and connections of various layers of the twin system and introduces the key technologies for implementing the framework, creating

parametric virtual models with Blender, training finite element models with artificial neural networks, establishing motion models, real-time monitoring with intelligent systems, and controlling jacking operations. Finally, using a BM from a specific project as an example, a DT system for monitoring the operational status of BM jacking was established, demonstrating the system's functionality. This system achieved real-time mapping of the operational posture and online prediction of structural performance during jacking operations. The principal contributions of this paper are as follows:

1. The constructed DT system achieves quantitative monitoring, evaluation, early warning, real-time prediction of operational posture, and remote control of the BMJOs, changing the previous separation of monitoring and control.
2. The prediction results are accurate, and the calculations are fast, thereby improving the safety assurance capabilities and the level of informatization and intelligence of BMJOs, effectively reducing safety risks and accidents in super high-rise construction projects.
3. The proposed DT system for monitoring the operational status of BM jacking not only achieves online prediction of the structural performance of jacking operations but can also be extended to other complex construction equipment and scenarios, such as tower cranes, climbing frames, and lifting equipment. By establishing corresponding motion models and monitoring systems, it can achieve real-time prediction and optimized control. In large bridge and tunnel construction, the proposed framework method can monitor structural changes and stress distribution during the construction process, thus predicting potential risks. In high-rise construction, the proposed DT system can monitor the construction progress, structural health, and environmental impact of high-rise buildings, enhancing construction efficiency and safety.

However, the DT construction method proposed in this paper has several shortcomings:

1. The proposed DT method employs sensor data as input information for real-time structural performance prediction. The accuracy of the sensor data determines the correctness of the prediction results. Future research can explore the adoption of more advanced sensing technologies and data processing methods, such as deep learning algorithms for data correction and cleaning and image recognition technology to improve data collection accuracy. Additionally, integrating multi-source data can be explored to enhance the overall accuracy and reliability of the data.
2. The goal of DT technology is to achieve virtual–real integration, ultimately enabling virtual control of physical systems. In this study, the real-time predicted structural performance and motion model are only displayed on the platform without autonomous decision-making capabilities. Future research should aim to enhance the analytical and decision-making abilities of DT models by integrating AI and machine learning (ML) technologies to achieve autonomous decision-making and optimized control. Specific solutions include developing an autonomous decision-making system based on reinforcement learning, designing adaptive control algorithms, and achieving real-time adjustments and optimization of the BM.
3. As the prediction and motion models are continuously updated and maintained in real-time, the volume of generated information and data increases significantly, making appropriate data storage and cleaning methods urgent issues to address. Future research should focus on developing suitable data storage and management strategies, leveraging cloud computing and distributed storage technologies to effectively manage and clean large volumes of real-time data. Additionally, blockchain technology can be introduced to ensure data security and traceability. Designing a data lifecycle management system will optimize data storage and cleaning processes.

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Abbreviations

The following abbreviations are used in this manuscript:

BM	Building machine
BMJO	Building machine jacking operation
DT	Digital twin
IoT	Internet of Things
AI	Artificial intelligence
3D	Three-dimensional
BMJO-DT	Building machine jacking operation digital twin
QEM	Quadric error metric
RBF	Radial basis function
MSE	Mean squared error
ICM	Local industrial control computer
PLC	Programmable logic controller
PID	Proportional-integral-derivative
ML	Machine learning

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